

Course Code: 2059 | Semester: 2 | Diploma in Automobile Engineering

Course Overview

Objectives

- Hands-on experience with basic automobile hand tools.
- Identification of engine components through dismantling.
- Understanding basic engine experiments and servicing procedures.
- Enhancing practical automotive servicing skills.

Teaching Scheme

Category	Engineering Science
Periods/Week	3 (Practical)
Total Periods	45
Credits	No Credit (Audit/Lab focus)

Official Source Declaration

This lesson is strictly mapped to the **SITTTR Kerala Revision 2021** syllabus for Course Code 2059. All modules, outcomes, and topics are derived from the official

Course Outcomes (COs)

CO No.	Description	Hours	Cognitive Level
CO1	Demonstrate the use of different tools used in Automobile workshop	7	Applying
CO2	Identify the application of measuring tools and specific tools in Automobile Garage.	7	Applying
CO3	Identify various engine components in an Automobile after dismantling.	14	Applying
CO4	Apply the procedure of maintenance schedule on two and four-wheeler vehicles.	14	Applying

Module 1: Automobile Workshop Tools

Duration: 7 Hours | CO1

M1.01: Identification of Hand Tools

Automobile servicing requires a variety of specialized hand tools to ensure fasteners are handled without damage and components are removed safely.

- **Spanners:** Open-end, Ring, and Combination spanners. Used for tightening/loosening bolts.
- **Socket Sets:** Used with ratchets for faster operation in confined spaces.
- **Wrenches:** Adjustable wrenches, Pipe wrenches, and Torque wrenches.
- **Pliers:** Combination pliers, Nose pliers, and Circlip pliers (Internal/External).
- **Hammers:** Ball peen, Cross peen, and Soft-faced (Mallet) hammers.
- **Screwdrivers:** Flat head, Phillips head, and Torx drivers.

மாண்புமிகு கல்வி அமைச்சர்: தொழில்நுட்ப கல்வியை மேம்படுத்தும் வகையில், தொழில்நுட்ப கல்வி மையங்களை, தொழில்நுட்ப கல்வி மையங்களை, தொழில்நுட்ப கல்வி மையங்களை

M1.02: Special Tools and Competency

Special tools are designed for specific tasks that standard tools cannot perform efficiently.

- **Torque Wrench:** Essential for tightening cylinder head bolts to specific manufacturer values to prevent warping.
- **Feeler Gauge:** Used to measure small gaps, such as valve clearance (tappet setting) or spark plug gaps.
- **Pullers:** Gear pullers and bearing pullers used to remove components pressed onto shafts.

Module 2: Measuring Tools & Engine Calculations

Duration: 7 Hours | CO2

M2.01 & M2.02: Bore and Stroke Measurement

Precision measurement is vital for determining engine wear and calculating displacement.

- **Bore Gauge:** A telescopic tool used to measure the internal diameter (Bore) of a cylinder. It helps identify "ovality" and "taper" wear.
- **Stroke Length:** The distance the piston travels from Top Dead Center (TDC) to Bottom Dead Center (BDC). It is exactly twice the **Crank Radius (r)**.

Stroke (L) = 2 × Crank Radius (r)

M2.03 & M2.04: Swept Volume and Compression Ratio

These calculations define the engine's capacity and thermal efficiency.

Swept Volume (V_s): The volume displaced by the piston during one stroke.

Formula: $V_s = (\pi/4) \times D^2 \times L$

Where D = Bore diameter, L = Stroke length.

Compression Ratio (CR): The ratio of total cylinder volume to clearance volume.

Formula: $CR = (V_s + V_c) / V_c$

Where V_c = Clearance Volume (volume above piston at TDC).

Formula Bank & Interactive Calculator

Bore (D) in mm:

e.g. 70

Stroke (L) in mm:

e.g. 80

Clearance Vol (V_c) in cc:

e.g. 40

Calculate

Visual Library

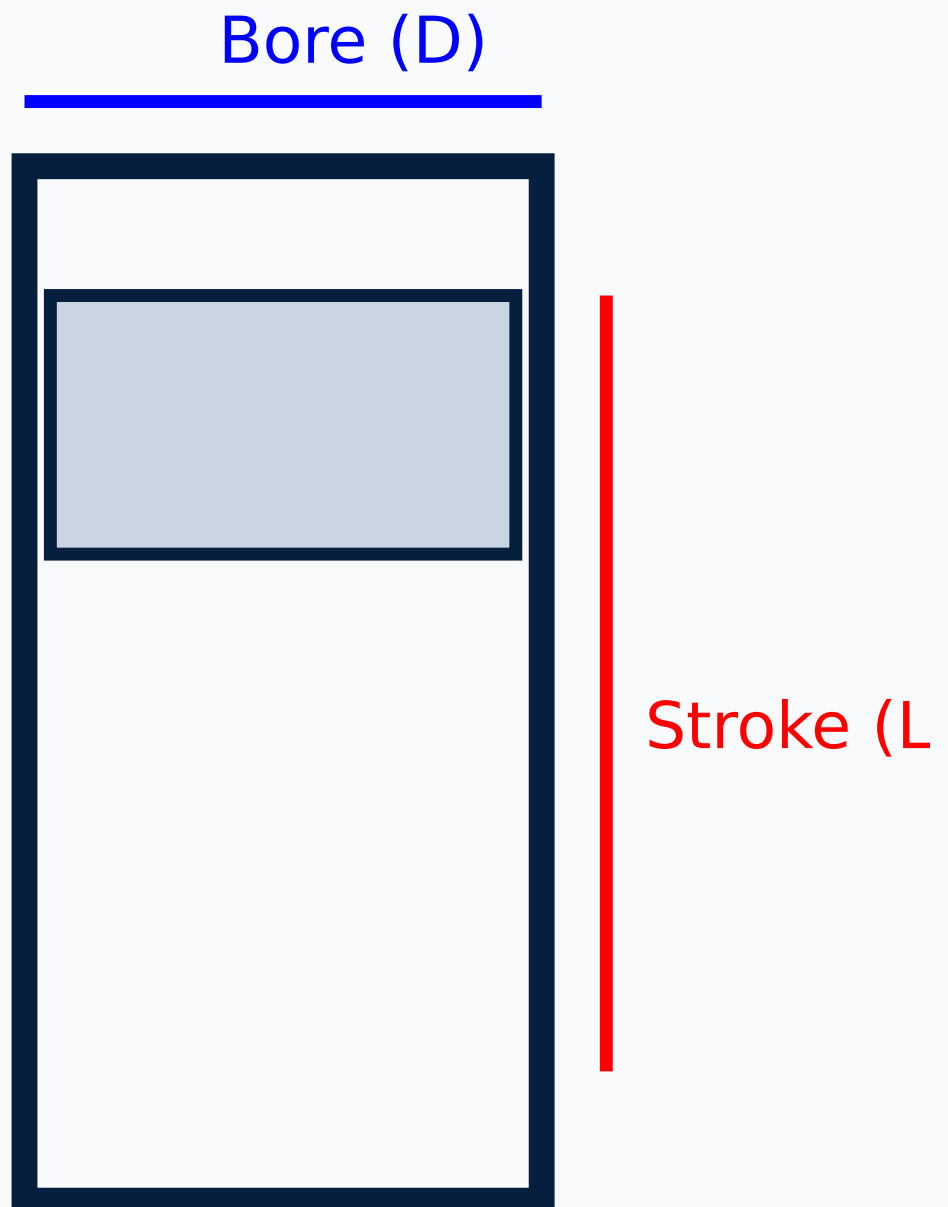


Figure 1: Basic Engine Cylinder Dimensions (Bore and Stroke)

Module 3: Engine Components Identification

Duration: 14 Hours | CO3

This module involves the actual dismantling of an engine to identify internal parts.

Component	Function / Key Details
Cylinder Block	The main structure of the engine; contains cylinders.
Cylinder Liners	Dry Liner: No direct contact with coolant. Wet Liner: Direct contact with coolant.
Piston & Rings	Transmits force to con-rod. Includes Compression rings (seal gas) and Oil rings (scrape oil).
Gudgeon Pin	Connects the piston to the small end of the connecting rod.
Crankshaft	Converts reciprocating motion of piston into rotary motion.
Camshaft	Controls the opening and closing of valves.

1. **Engine Components:** The engine is composed of several major parts, including the cylinder block, cylinder liners, piston and rings, gudgeon pin, crankshaft, and camshaft. Each component has a specific function in the engine's operation.

Module 4: Maintenance & Servicing

Duration: 14 Hours | CO4

M4.01 - M4.04: Fluids and General Maintenance

- **Engine Oil:** Check level using dipstick. Identify viscosity (e.g., 20W40).
- **Coolant:** Check reservoir levels. Never open a hot radiator cap.
- **Jacking:** Use hydraulic or mechanical jacks at designated chassis points for wheel removal.
- **Greasing:** Applying lubricant to suspension joints, steering linkages, and propeller shaft universal joints.

M4.05 - M4.07: Spark Plug and Injector Testing

Spark Plug Servicing:

1. Remove plug using a spark plug socket.
2. Clean carbon deposits using a wire brush or plug cleaner.
3. Check gap using a feeler gauge (typically 0.6mm to 1.0mm).
4. Adjust gap by bending the side electrode.

Fuel Injector: Testing injection pressure and spray pattern using an injector tester (nozzle tester).

Practice Questions & Answers

1. What is the difference between a wet liner and a dry liner?

A wet liner comes into direct contact with the engine coolant, requiring seals at the top and bottom. A dry liner is pressed into the cylinder block and does not touch the coolant directly.

2. Define Compression Ratio.

Compression Ratio is the ratio of the total volume of the cylinder (Swept Volume + Clearance Volume) to the Clearance Volume.

3. Why is a torque wrench used for cylinder head bolts?

To ensure all bolts are tightened with equal and specific force as per manufacturer specs, preventing cylinder head warping and gasket failure.

Practice Model Lab Exam

Time: 3 Hours | Max Marks: 100

Part A: Identification (20 Marks)

Identify 5 engine components placed on the table and state their functions.

Part B: Measurement (30 Marks)

Measure the bore and stroke of the given engine. Calculate the swept volume.

Part C: Maintenance Task (50 Marks)

Perform spark plug cleaning and gap setting OR Demonstrate the procedure for engine oil level check and jacking up a vehicle.

Quick Revision

- Stroke** = $2 \times \text{Crank Radius}$.
- Swept Volume** is the capacity of one cylinder.
- Feeler Gauge** is for gaps; **Bore Gauge** is for internal diameters.
- Compression Rings** prevent blow-by; **Oil Rings** control lubrication.
- Always use **Safety Stands** after jacking up a vehicle.

Glossary

TDC	Top Dead Center (Highest point of piston travel).
BDC	Bottom Dead Center (Lowest point of piston travel).
Ovality	Difference in cylinder diameter measured in two perpendicular directions.
Viscosity	Resistance of oil to flow.

മലയാളം പിന്തുണ (Malayalam Support)

ഈ ഭാഗം മലയാളത്തിൽ വിവരങ്ങൾ നൽകുന്നു:

- പദപരിഭാഷ:** സൂചകം: സൂചകം (Bore), സൂചകം (Stroke)
- സൂചകം:** സൂചകം (Bore), സൂചകം (Stroke)

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3. **பொருள்:** பொருள் பொருளாக பொருள் பொருள் பொருளாக, பொருளாக பொருளாக, பொருளாக பொருளாக பொருளாகப்பெற்று.

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Reference: Basic Automobile Engineering - C P Nakra | Practical Automobile Engineering – Malhotra

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